

Mathematics: founded on Set Theory

O Chansol & the Echoes

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1 Foundations

1.1 Set Theory

0 Prerequisites of set theory

This theory builds on the theory developed in [Logic and Language](#).

1 In Preparation

1.0 Prerequisites of in preparation

This chapter builds on the theory developed in [Logic and Language](#).

1.1 An object being an element of a set defining axiom for a relation symbol

2 Algebra

2.1 Basic Algebra

0 Prerequisites of basic algebra

This theory builds on the theory developed in [Logic and Language](#), and [Set Theory](#).

1 In Preparation

1.0 Prerequisites of in preparation

This chapter builds on the theory developed in [Logic and Language](#), and [Set Theory](#).

1.1 An object being a magma defining axiom for a relation symbol

1.2 The binary operation of a magma defining axiom for a function symbol

1.3 A binary operation being associative on a set defining axiom for a relation symbol

1.4 The underlying set of a magma defining axiom for a function symbol

1.5 An object being an identity element with respect to a binary operation of a set defining axiom for a relation symbol

1.6 An object being an inverse element of an element with respect to a binary operation of a set defining axiom for a relation symbol

2.2 Group Theory

0 Prerequisites of group theory

This chapter builds on the theory developed in [Logic and Language](#), [Set Theory](#), and [Basic Algebra](#).

1 Groups

1.0 Prerequisites of groups

This chapter builds on the theory developed in [Logic and Language](#), [Set Theory](#), and [Basic Algebra](#).

1.1 A magma being a group defining axiom for a relation symbol

$$\begin{aligned} & \forall M((M \text{isA Magma}) \supset \\ & ((M \text{isA Group}) \diamond \\ & (((\text{the Binary Operation Of } M) \text{is Associative On } (\text{the Underlying Set Of } M)) \wedge \\ & \exists a(a \text{is An Identity Element With Respect To } (\text{the Binary Operation Of } M) \text{Of } (\text{the Underlying Set Of } M))) \wedge \\ & \forall a((a \text{is An Element Of } (\text{the Underlying Set Of } M)) \supset \exists b(b \text{is An Inverse Element Of } a \text{ With Respect To } (\text{the Binary Operation Of } M) \text{Of } (\text{the Underlying Set Of } M)))))) \end{aligned}$$